Anjuman-I-Islam's

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DEPARTMENT OF INFORMATION TECHNOLOGY

ITP701 MAJOR PROJECT - 1

Title of the Project

Blood Banking Via Cloud Computing

Supervisor/Guide

ER SWATI GURAV

REV - 2019 'C' Scheme

University of Mumbai

Academic Year (2026 -27)

CERTIFICATE

This is to certify that the project entitled "**Blood Bank Via Cloud Computing"** by **"Yamin Khan" (242464), " Mohammed Rumaan Shaikh" (242469)**, **"Abdurrahman Qureshi " (242466)**, **"Abdul Sameer Shaikh" (242471)** " submitted to the University of Mumbai in partial fulfilment of the requirement for the ITP701 Major Project - 1 for the 7TH Semester in **Department of Information Technology**.

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MAJOR PROJECT CO-ORDINATOR

(Dr. Zainab Mirza)
HEAD OF DEPARTMENT

MINI PROJECT REPORT APPROVAL

This project report entitled " **Blood Bank Via Cloud Computing** " by "**Yamin Khan" (242464),** "**Mohammed Rumaan Shaikh" (242469)**, "**Abdurrahman Qureshi " (242466)**, "**Abdul Sameer Shaikh" (242471)** is approved for the ITP701 Major Project 1 for project of the 7th Semester.

EXAMINER 1

EXAMINER 2

Date :

Place : Mumbai

ACKNOWLEDGEMENT

We would take this humble opportunity to express our sincere gratitude to our project guide ER. Swati Gurav for her valuable advice and direction under

Which the project was executed and to all respected people who provided their kind support and encouragement throughout the project. The success of this project to the most extent rests on every bit of suggestions, advice, and help which they provided us from time to time and also being open and constructive to all the problems that we faced during the entire development lifecycle of our project.

We would also like to extend a big vote of thanks to the coordinator, our teachers, for their constant support and encouragement

YAMIN KHAN (242464)

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ABSTRACT

The Blood Banking Via Cloud Computing project is designed to provide a centralized, efficient, and easily accessible platform for managing blood-bank activities using cloud computing technology. Traditional blood-bank systems often depend on manual records or isolated databases, which can make it difficult to obtain updated information about donors, blood groups, blood availability, and blood requests.

The proposed system provides a digital platform where **donors, hospitals/recipients, and administrators** can interact with the blood-bank system. Donors can register and maintain their information, while authorized users can check blood availability and submit blood requests. The administrator can manage donor records, blood-stock information, requests, and other system activities.

Cloud computing provides centralized data storage, accessibility, scalability, and easier management of the system. The project aims to reduce manual work, improve data accessibility, minimize record-management errors, and provide faster access to blood-related information. The system can also be extended in the future with features such as notifications, location-based blood-bank searching, analytics, and emergency blood requests.

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INTRODUCTION

Blood is one of the most important resources required during medical emergencies, surgeries, accidents, and treatment of various diseases. Blood banks are responsible for collecting, storing, managing, and distributing blood according to the requirements of patients and healthcare organizations.

In a conventional blood-bank system, information about donors, blood groups, stock, and requests may be maintained through paperwork or separate computer systems. Such approaches can result in **data duplication, difficulty in updating records, delayed information retrieval, and limited accessibility.**

Cloud computing provides an effective solution to these problems by allowing application data and services to be hosted on centralized cloud infrastructure. Authorized users can access the system through the internet without depending on a particular physical computer.

The proposed **Blood Banking Via Cloud Computing** system aims to create a centralized platform for managing important blood-bank operations. The system can maintain information about:

- Blood donors
- Blood groups
- Available blood units
- Blood collection and storage
- Blood requests
- Hospitals/recipients
- User accounts
- Administrative activities

The administrator acts as the main controller of the system and can verify and manage information. Donors and authorized users can access relevant services through the application.

Objectives

The major objectives of the project are:

1. To develop a centralized cloud-based blood-bank management system.
2. To maintain donor information digitally.
3. To maintain and update blood-stock information.
4. To allow users to check blood availability.
5. To provide a facility for submitting blood requests.
6. To reduce dependency on manual record keeping.

7. To improve the accessibility and management of blood-bank information.
8. To provide secure authentication and controlled access.
9. To reduce errors and duplication in blood-bank records.
10. To provide a scalable platform that can be enhanced in the future.

METHODOLOGY

The project follows a systematic development methodology consisting of **requirement analysis, research, system design, technology planning, implementation, testing, and evaluation**.

Step 1: Requirement Analysis

The requirements of different users are identified.

Donor requirements:

- Registration and login
- Profile management
- Blood-group information
- Donation-related information

Recipient/Hospital requirements:

- Login
- Search/check blood availability
- Submit blood requests
- Track request status

Administrator requirements:

- Manage donors
- Manage blood inventory
- Manage requests
- Manage users
- Update system information

Step 2: Literature Review

Existing blood-bank management systems and cloud-based healthcare applications are studied. Their features, advantages, and limitations are analyzed to identify possible improvements.

Step 3: Research Gap Identification

The limitations of existing systems are identified, such as:

- Manual record maintenance
- Limited accessibility
- Delayed stock updates
- Difficulty in maintaining centralized information
- Lack of scalable infrastructure

Step 4: System Design

The overall system architecture, database structure, user roles, data flow, and application workflow are designed.

Step 5: Technology Planning

A suitable technology stack is selected based on project requirements. A possible implementation can use:

Layer	Technologies
Frontend	NextJS, ShadCN, React.js, React Native, Flutter
Backend	Node.js, Express.js
Database	MongoDB Atlas
Cloud	AWS (EC2, S3, Monitoring), Vercel
API Testing	Postman, Jest, Supertest
Authentication and Security	Clerk, JWT, bcrypt, HTTPS, RBAC, Audit Logging
Maps and Notifications	Google Maps API, Firebase Cloud Messaging
Dev Tools	Git, GitHub, Postman, VS Code, NPM
Code Quality	ESLint, Prettier,

Step 6: Implementation

The application is developed module by module:

1. User authentication
2. Donor registration
3. Donor profile
4. Blood inventory
5. Blood search
6. Blood request
7. Administrator dashboard
8. Database integration

Step 7: Testing

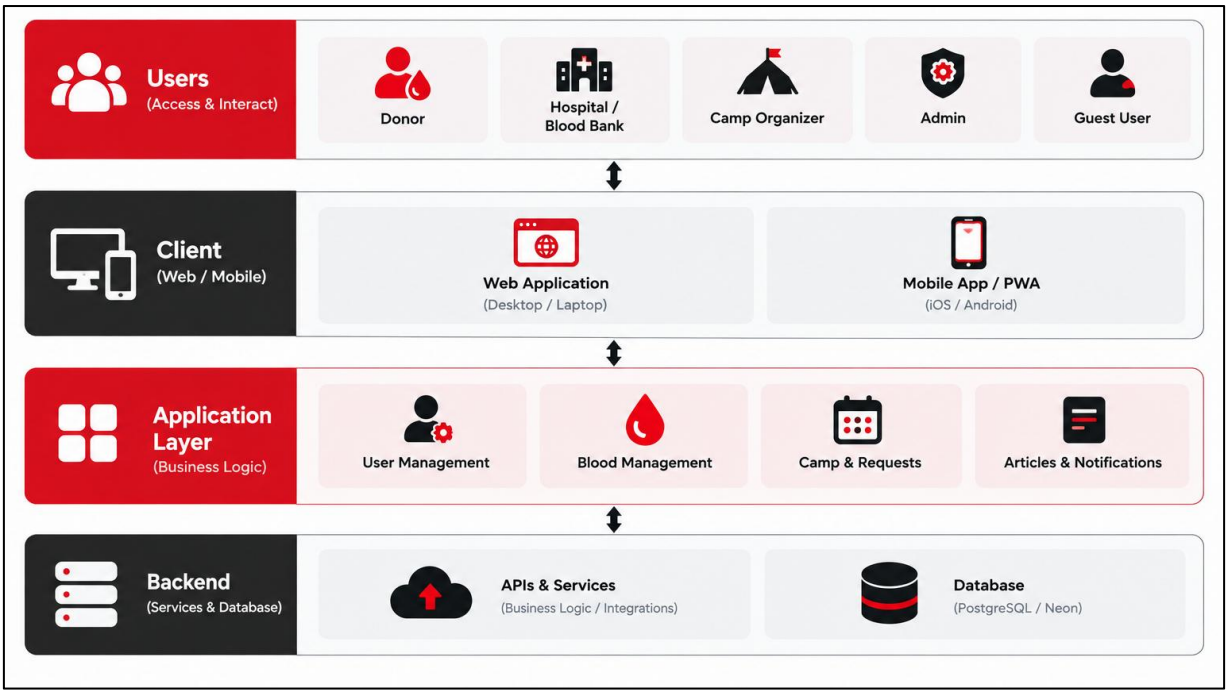
Individual modules are tested first, followed by integration testing and complete system testing.

Step 8: Deployment and Evaluation

After successful testing, the application can be deployed on cloud infrastructure. Its performance, usability, reliability, and functionality are evaluated.

IMPLEMENTATION AND DESIGN

System Architecture



Working of the Architecture

- Users access the blood-bank application through the web interface.
- The presentation layer accepts user input.
- Requests are sent to the application/backend layer.
- The application layer validates the request and performs the required operation.
- Relevant information is retrieved from or stored in the cloud database.
- The result is sent back to the user interface.
- Cloud infrastructure provides centralized storage, availability, scalability, and backup facilities.

System Modules

1 User Authentication Module

This module provides:

- User registration
- Login
- Logout
- Password validation
- Role-based access
- Authentication of authorized users

2 Donor Management Module

The donor module maintains:

- Donor name
- Contact details
- Blood group
- Age and basic information
- Donation history
- Donor availability/status

3 Blood Inventory Module

This module manages:

- Blood groups
- Available blood units
- Stock addition
- Stock reduction
- Blood availability
- Inventory updates

4 Blood Request Module

Users or hospitals can:

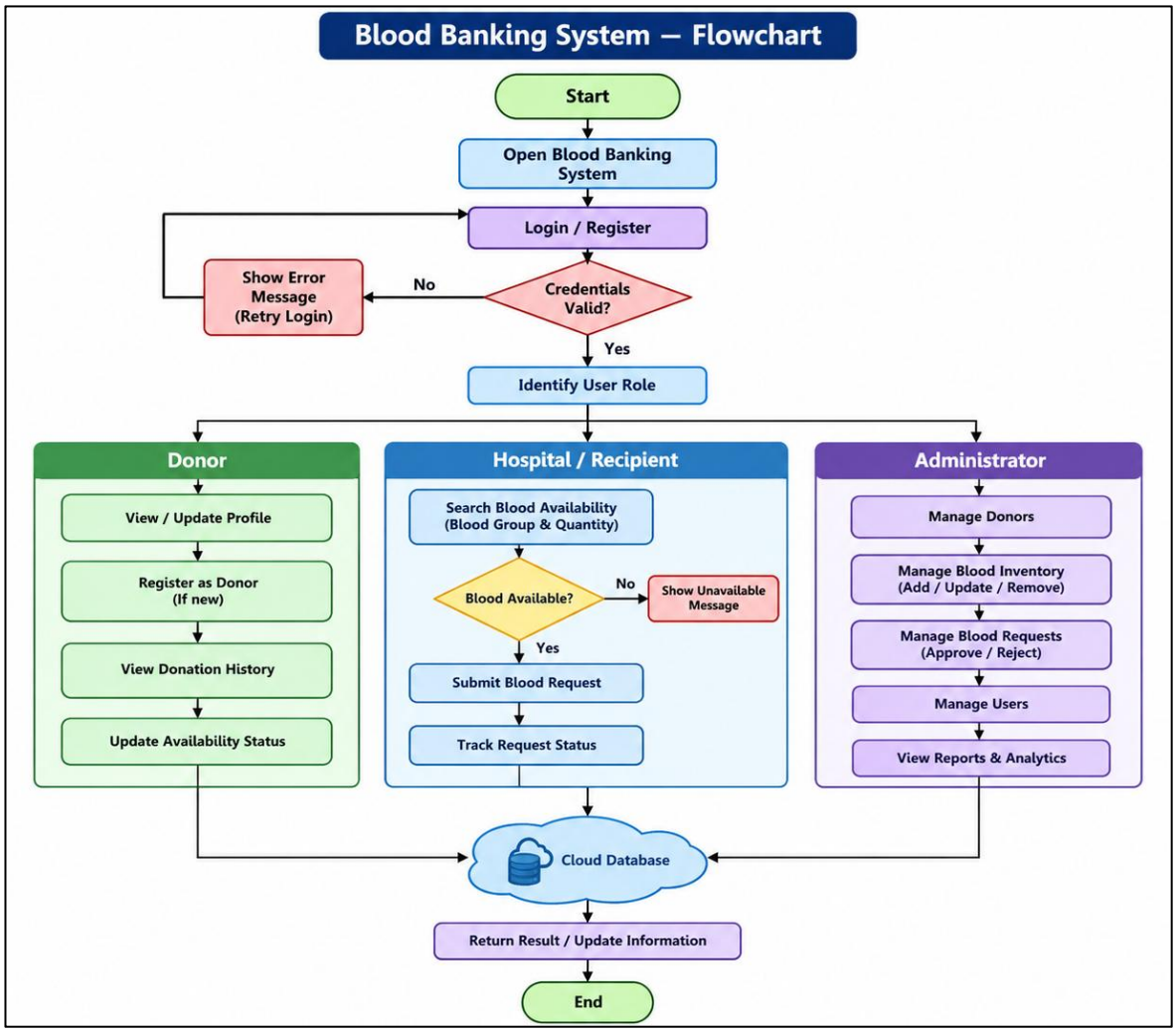
- Search for required blood
- Submit blood requests
- Specify blood group and quantity
- View request status
- Receive information about request processing

5 Administrator Module

The administrator can:

- Manage users
- Verify donor information
- Manage blood stock
- Manage requests
- Update records
- Monitor system activities

Flowchart : (Overall)



TESTING /RESULT & ANAYLIS

Testing is performed to verify that each component of the Blood Banking Via Cloud Computing system works correctly and that different modules can communicate properly.

Types of Testing

1. Unit Testing

Individual modules such as login, registration, donor management, and blood inventory are tested independently.

2. Integration Testing

Different modules are tested together to ensure that data is transferred correctly between the frontend, backend, and database.

3. Functional Testing

The main functions of the system are tested according to the defined requirements.

4. Validation Testing

User inputs such as email, password, blood group, quantity, and required fields are validated.

5. Authentication Testing

Login credentials and access permissions are tested to ensure that unauthorized users cannot access restricted features.

6. Database Testing

Data insertion, retrieval, modification, and deletion operations are tested.

7. User Interface Testing

The interface is checked for proper navigation, readability, responsiveness, and ease of use.

Test Cases:

Test Case	Test Scenario	Expected Result	Result
TC01	User registration with valid data	Account should be created	Pass
TC02	Registration with missing fields	Validation message should appear	Pass
TC03	Login with valid credentials	User should be logged in	Pass
TC04	Login with incorrect password	Access should be denied	Pass
TC05	Donor enters valid information	Donor record should be stored	Pass
TC06	Search for available blood group	Available stock should be displayed	Pass
TC07	Search unavailable blood group	System should show unavailable status	Pass
TC08	Submit valid blood request	Request should be recorded	Pass
TC09	Admin updates blood stock	Updated stock should be displayed	Pass
TC10	Unauthorized user opens admin page	Access should be restricted	Pass
TC11	Logout	User session should end	Pass
TC12	Retrieve stored records	Correct records should be displayed	Pass

Result: Output Images



Fig.1

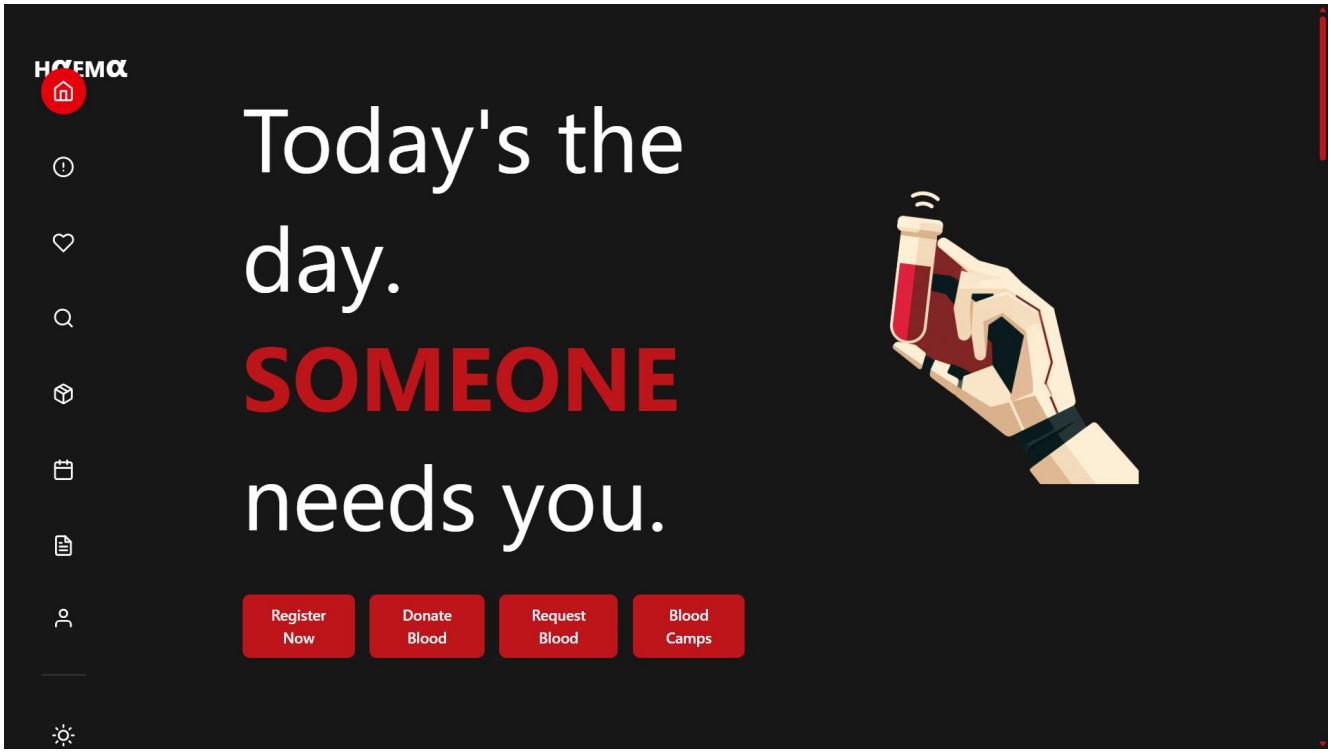

Today's the day.
SOMEONE needs you.
Register Now
Donate Blood
Request Blood
Blood Camps

Fig.2

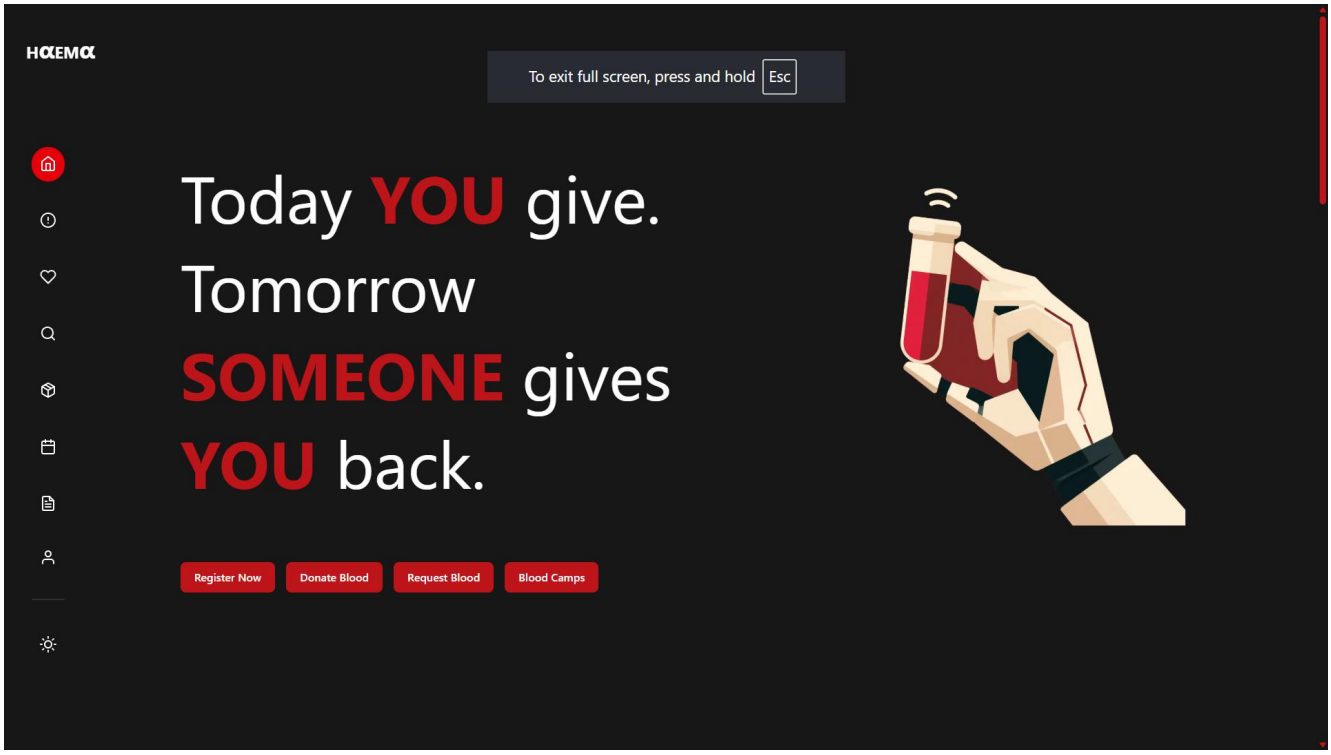

To exit full screen, press and hold Esc
Today YOU give. Tomorrow SOMEONE gives YOU back.
Register Now
Donate Blood
Request Blood
Blood Camps

Fig.3

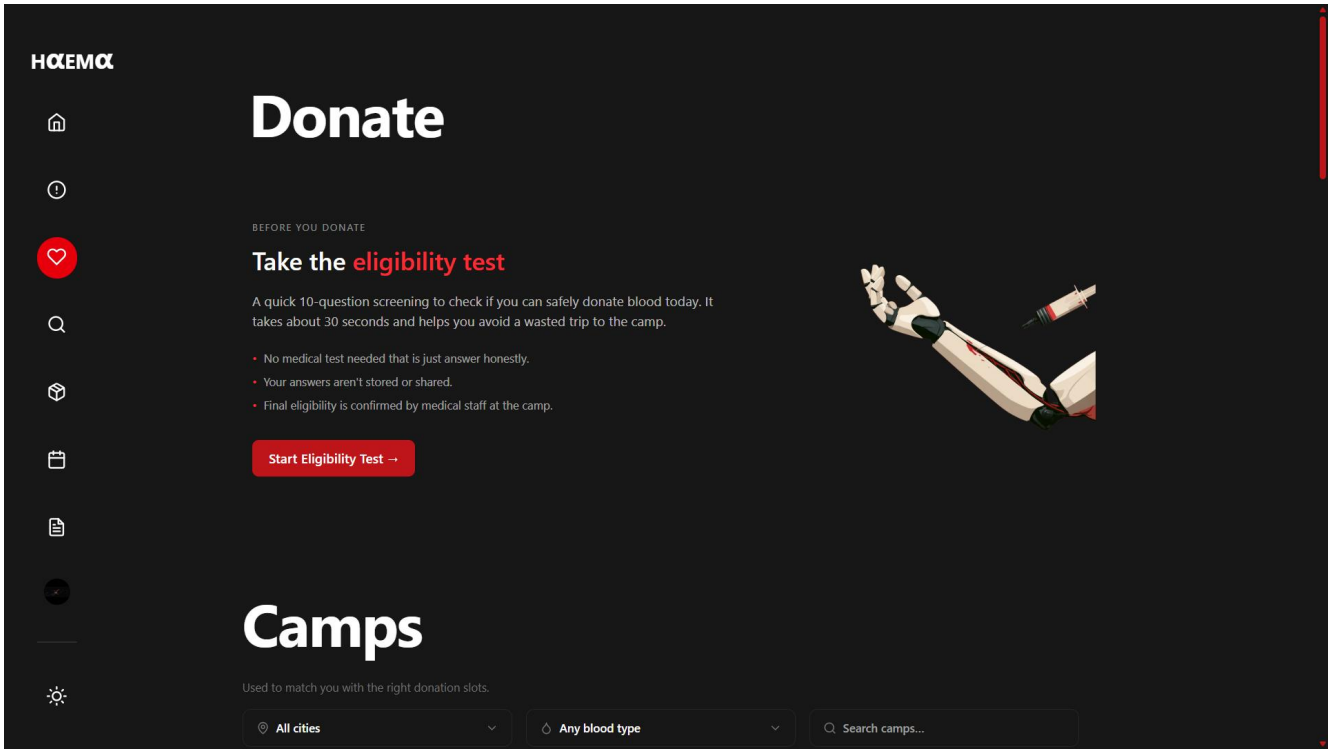

HαEMα
Donate
BEFORE YOU DONATE
Take the eligibility test
A quick 10-question screening to check if you can safely donate blood today. It takes about 30 seconds and helps you avoid a wasted trip to the camp.
No medical test needed that is just answer honestly.
Your answers aren't stored or shared.
Final eligibility is confirmed by medical staff at the camp.
Start Eligibility Test →
Camps
Used to match you with the right donation slots.
All cities
Any blood type
Search camps...

Fig.4

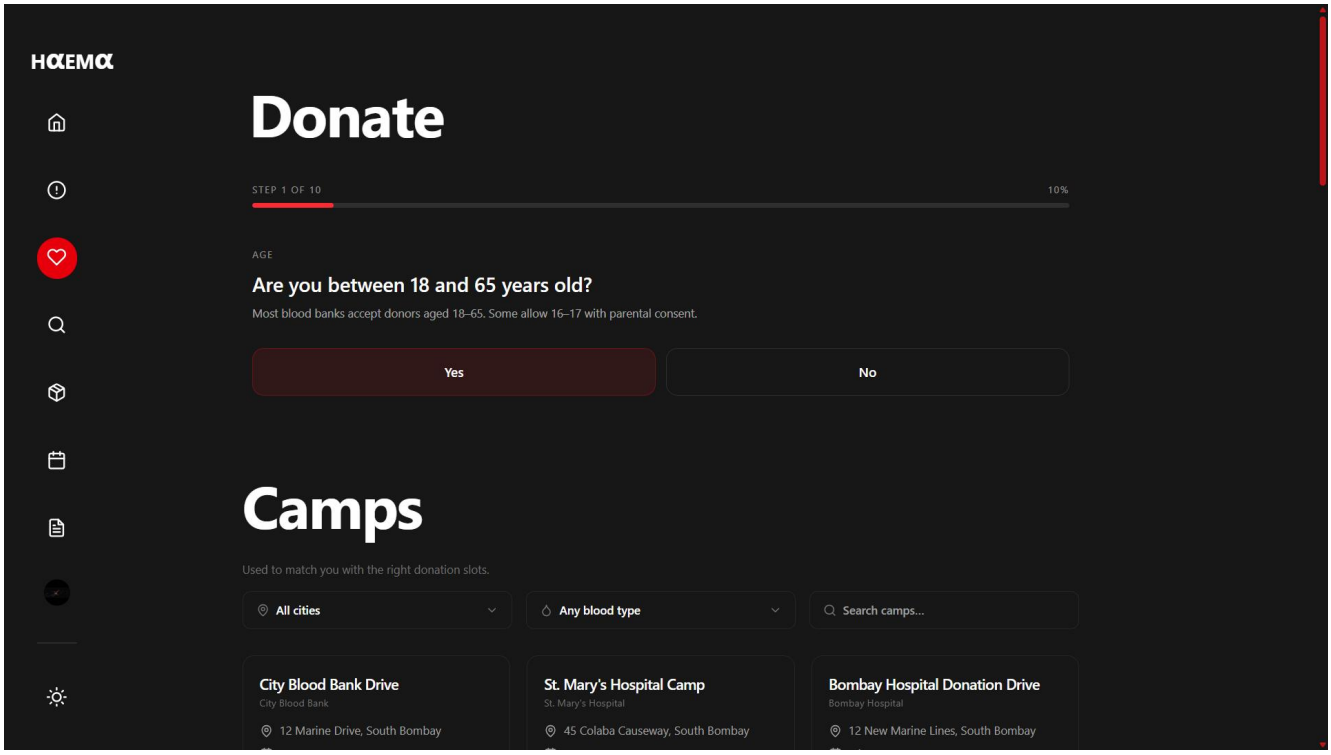

HαEMα
Donate
STEP 1 OF 10
10%
AGE
Are you between 18 and 65 years old?
Most blood banks accept donors aged 18–65. Some allow 16–17 with parental consent.
Yes
No
Camps
Used to match you with the right donation slots.
All cities
Any blood type
Search camps...
City Blood Bank Drive
City Blood Bank
12 Marine Drive, South Bombay
St. Mary's Hospital Camp
St. Mary's Hospital
45 Colaba Causeway, South Bombay
Bombay Hospital Donation Drive
Bombay Hospital
12 New Marine Lines, South Bombay

Fig.5

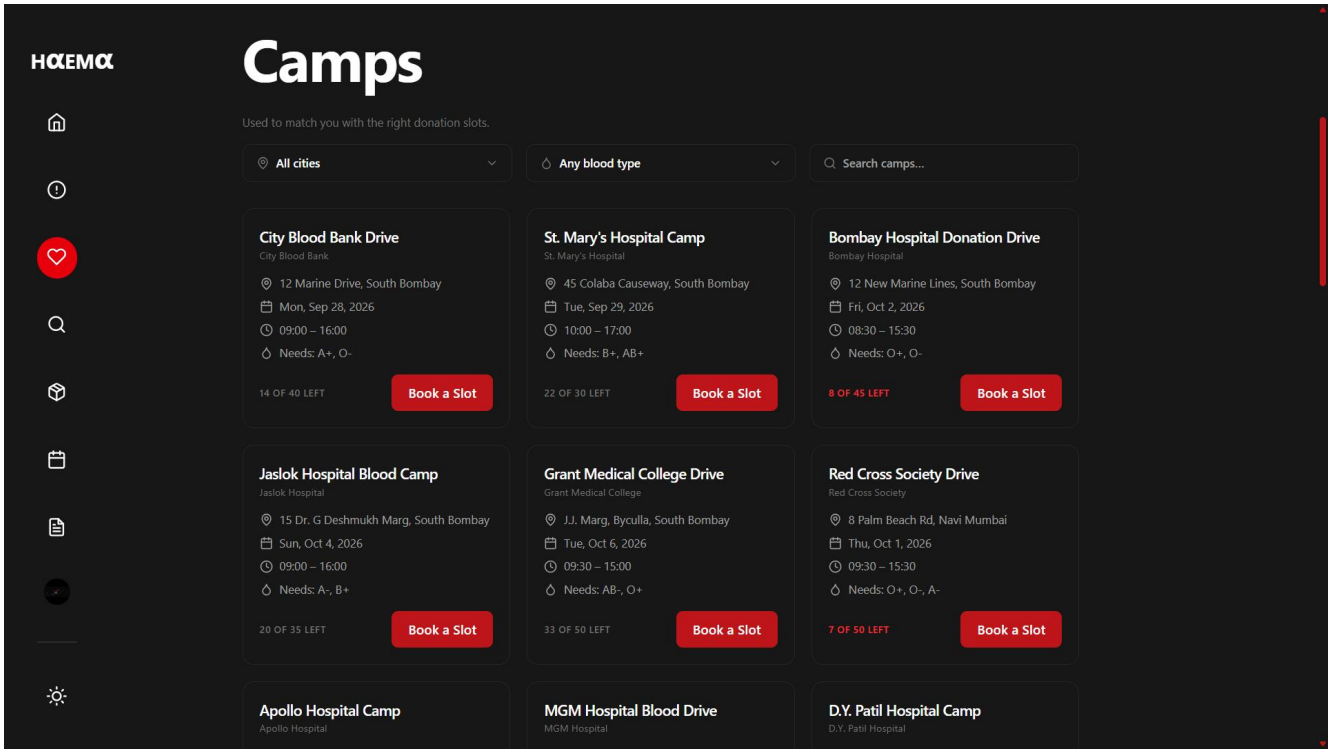

HαEMα
Camps
Used to match you with the right donation slots.
All cities
Any blood type
Search camps...
City Blood Bank Drive
City Blood Bank
12 Marine Drive, South Bombay
Mon, Sep 28, 2026
09:00 – 16:00
Needs: A+, O-
14 OF 40 LEFT
Book a Slot
St. Mary's Hospital Camp
St. Mary's Hospital
45 Colaba Causeway, South Bombay
Tue, Sep 29, 2026
10:00 – 17:00
Needs: B+, AB+
22 OF 30 LEFT
Book a Slot
Bombay Hospital Donation Drive
Bombay Hospital
12 New Marine Lines, South Bombay
Fri, Oct 2, 2026
08:30 – 15:30
Needs: O+, O-
8 OF 45 LEFT
Book a Slot
Jaslok Hospital Blood Camp
Jaslok Hospital
15 Dr. G Deshmukh Marg, South Bombay
Sun, Oct 4, 2026
09:00 – 16:00
Needs: A-, B+
20 OF 35 LEFT
Book a Slot
Grant Medical College Drive
Grant Medical College
JJ. Marg, Byculla, South Bombay
Tue, Oct 6, 2026
09:30 – 15:00
Needs: AB-, O+
33 OF 50 LEFT
Book a Slot
Red Cross Society Drive
Red Cross Society
8 Palm Beach Rd, Navi Mumbai
Thu, Oct 1, 2026
09:30 – 15:30
Needs: O+, O-, A-
7 OF 50 LEFT
Book a Slot
Apollo Hospital Camp
Apollo Hospital
MGM Hospital Blood Drive
MGM Hospital
D.Y. Patil Hospital Camp
D.Y. Patil Hospital

Fig.6

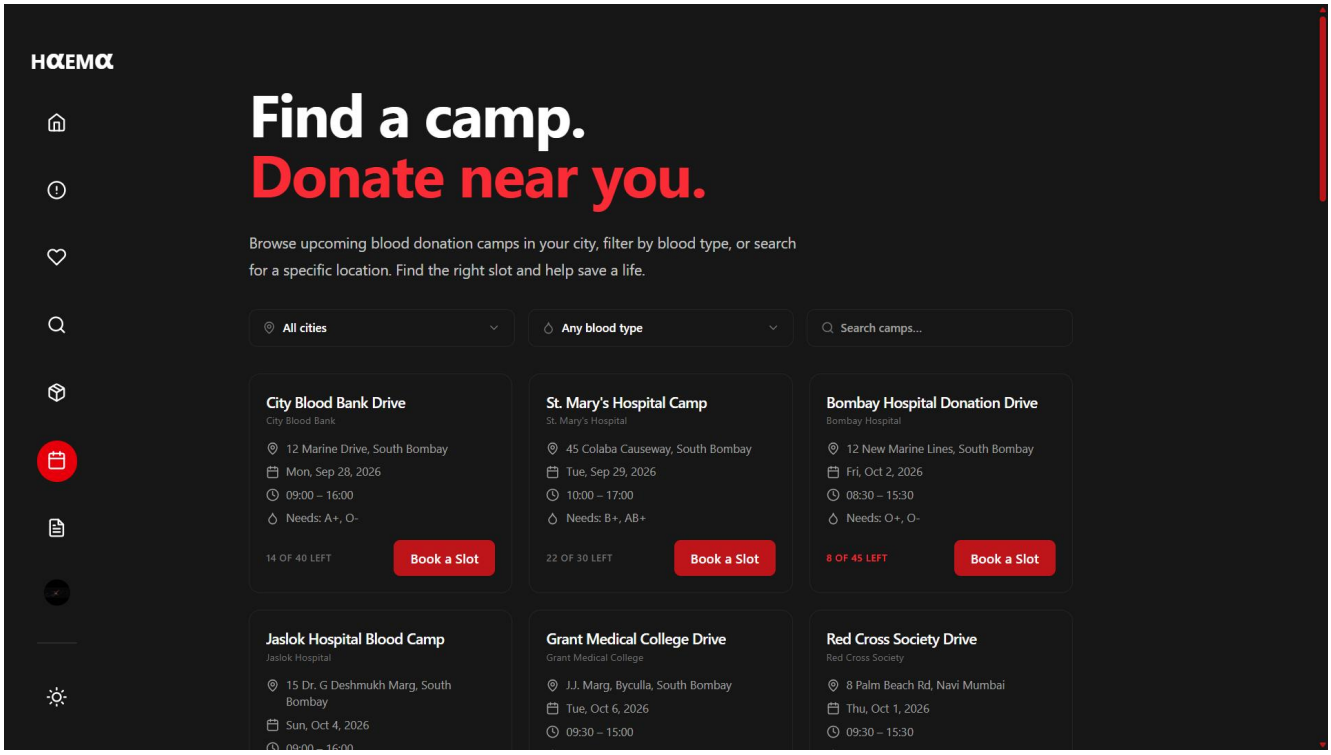

HαEMα
Find a camp.
Donate near you.
Browse upcoming blood donation camps in your city, filter by blood type, or search for a specific location. Find the right slot and help save a life.
All cities
Any blood type
Search camps...
City Blood Bank Drive
City Blood Bank
12 Marine Drive, South Bombay
Mon, Sep 28, 2026
09:00 – 16:00
Needs: A+, O-
14 OF 40 LEFT
Book a Slot
St. Mary's Hospital Camp
St. Mary's Hospital
45 Colaba Causeway, South Bombay
Tue, Sep 29, 2026
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Needs: B+, AB+
22 OF 30 LEFT
Book a Slot
Bombay Hospital Donation Drive
Bombay Hospital
12 New Marine Lines, South Bombay
Fri, Oct 2, 2026
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JJ. Marg, Byculla, South Bombay
Tue, Oct 6, 2026
09:30 – 15:00
Red Cross Society Drive
Red Cross Society
8 Palm Beach Rd, Navi Mumbai
Thu, Oct 1, 2026
09:30 – 15:30

Fig.7

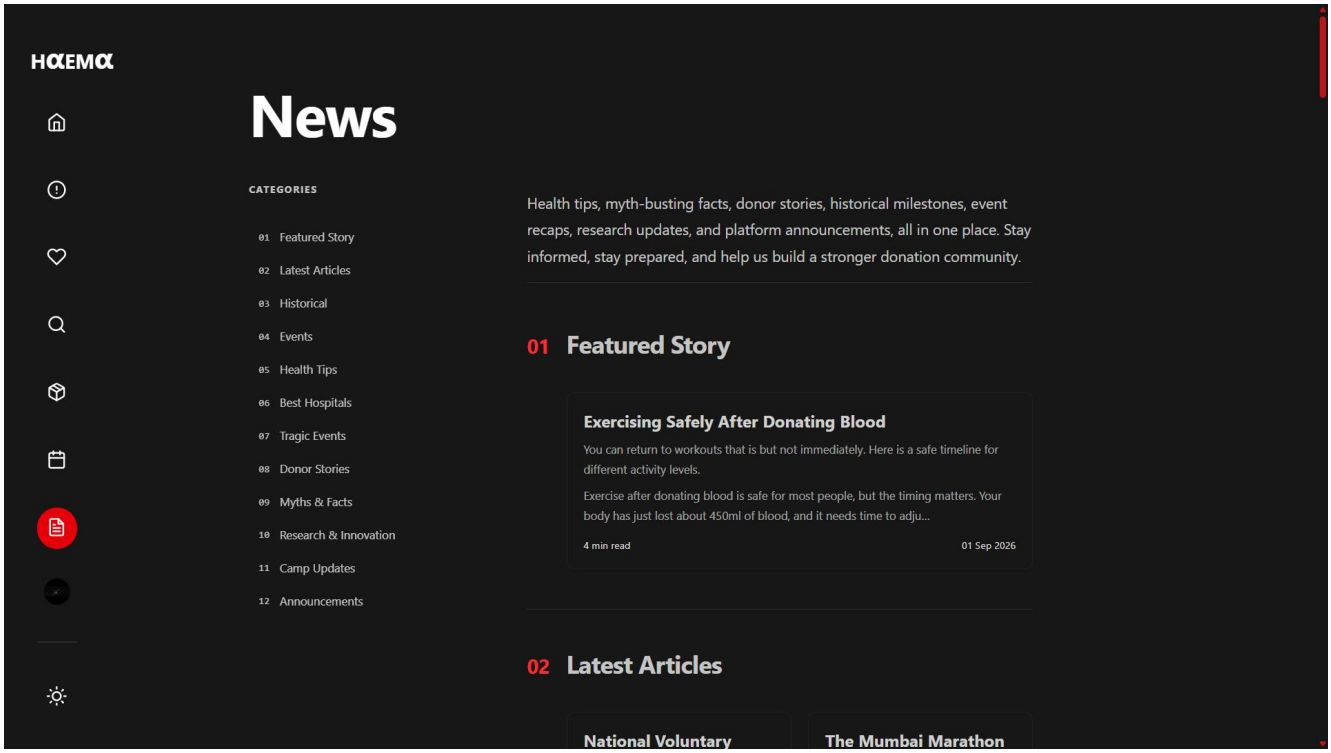

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News
CATEGORIES
01 Featured Story
02 Latest Articles
03 Historical
04 Events
05 Health Tips
06 Best Hospitals
07 Tragic Events
08 Donor Stories
09 Myths & Facts
10 Research & Innovation
11 Camp Updates
12 Announcements
Health tips, myth-busting facts, donor stories, historical milestones, event recaps, research updates, and platform announcements, all in one place. Stay informed, stay prepared, and help us build a stronger donation community.
01 Featured Story
Exercising Safely After Donating Blood
You can return to workouts that is but not immediately. Here is a safe timeline for different activity levels.
Exercise after donating blood is safe for most people, but the timing matters. Your body has just lost about 450ml of blood, and it needs time to adju...
4 min read
01 Sep 2026
02 Latest Articles
National Voluntary
The Mumbai Marathon

Fig.8

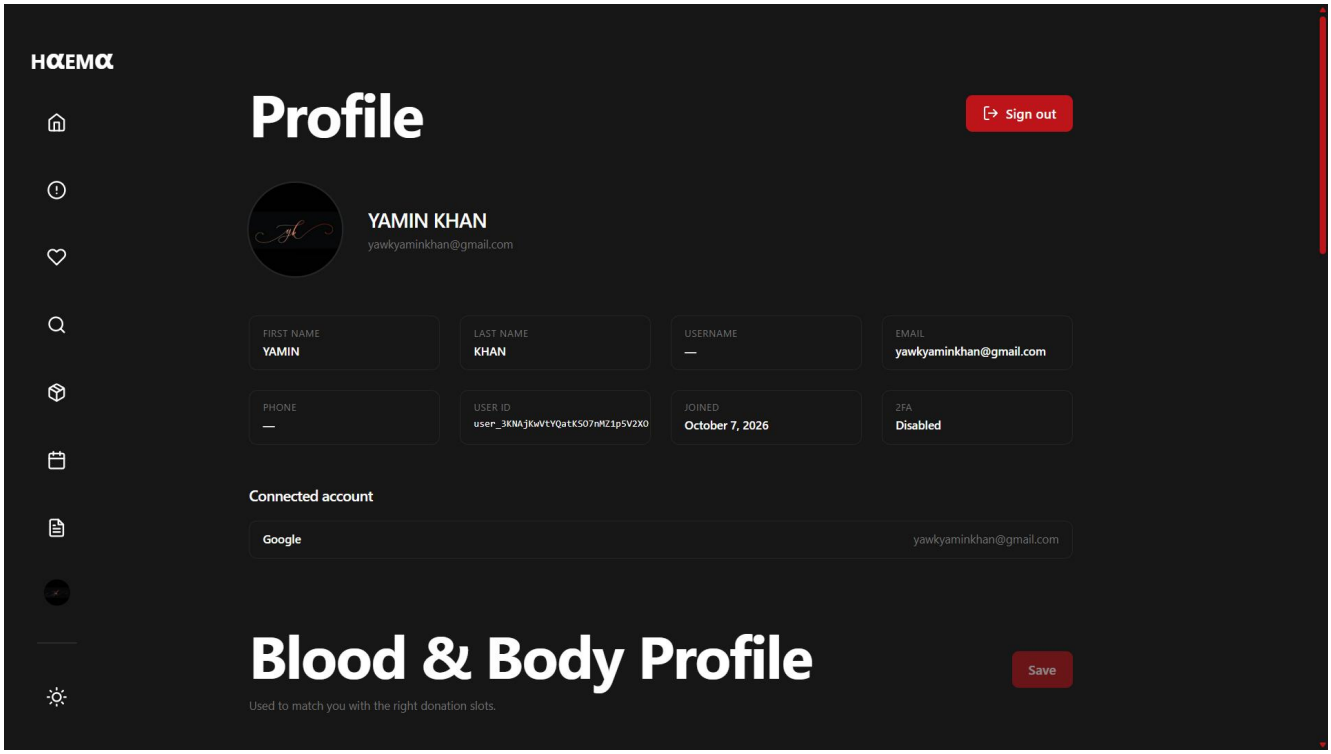

HαEMα
Profile
Sign out
YAMIN KHAN
yawkyaminkhan@gmail.com
FIRST NAME
YAMIN
LAST NAME
KHAN
USERNAME
—
EMAIL
yawkyaminkhan@gmail.com
PHONE
—
USER ID
user_3KNAjKwVtYQatKSO7nMZ1p5V2XO
JOINED
October 7, 2026
2FA
Disabled
Connected account
Google
yawkyaminkhan@gmail.com
Blood & Body Profile
Used to match you with the right donation slots.
Save

Fig.9

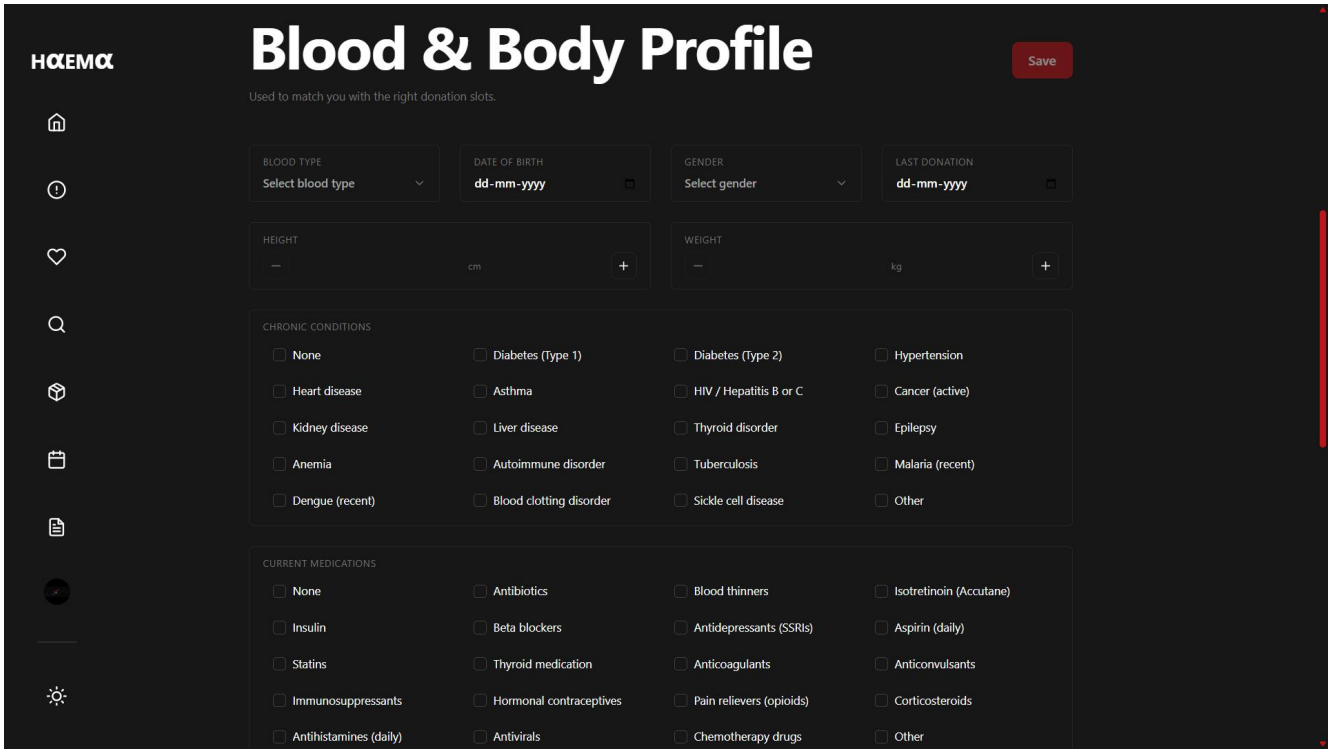

HAEMA
Blood & Body Profile
Save
Used to match you with the right donation slots.
BLOOD TYPE
Select blood type
DATE OF BIRTH
dd-mm-yyyy
GENDER
Select gender
LAST DONATION
dd-mm-yyyy
HEIGHT
cm
WEIGHT
kg
CHRONIC CONDITIONS
None
Diabetes (Type 1)
Diabetes (Type 2)
Hypertension
Heart disease
Asthma
HIV / Hepatitis B or C
Cancer (active)
Kidney disease
Liver disease
Thyroid disorder
Epilepsy
Anemia
Autoimmune disorder
Tuberculosis
Malaria (recent)
Dengue (recent)
Blood clotting disorder
Sickle cell disease
Other
CURRENT MEDICATIONS
None
Antibiotics
Blood thinners
Isotretinoin (Accutane)
Insulin
Beta blockers
Antidepressants (SSRIs)
Aspirin (daily)
Statins
Thyroid medication
Anticoagulants
Anticonvulsants
Immunosuppressants
Hormonal contraceptives
Pain relievers (opioids)
Corticosteroids
Antihistamines (daily)
Antivirals
Chemotherapy drugs
Other

Fig.10

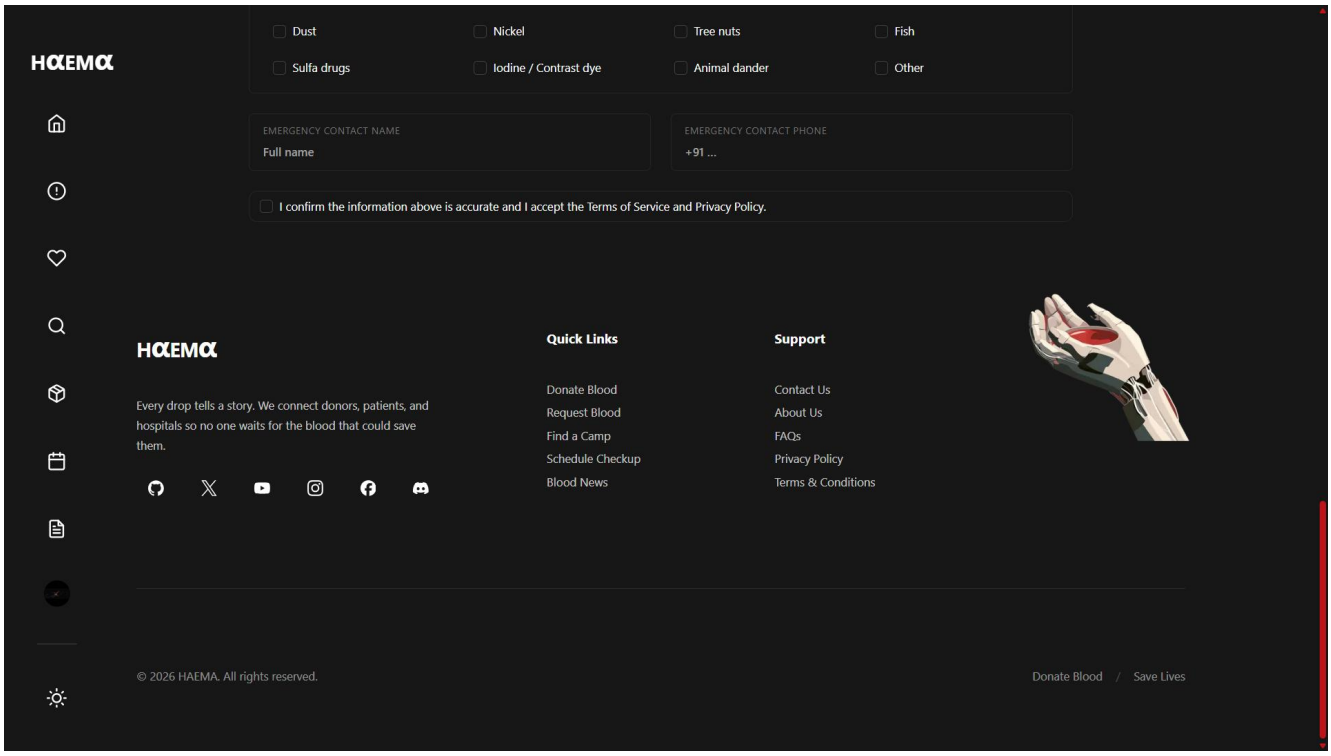

HAEMA
Dust
Nickel
Tree nuts
Fish
Sulfa drugs
Iodine / Contrast dye
Animal dander
Other
EMERGENCY CONTACT NAME
Full name
EMERGENCY CONTACT PHONE
+91 ...
I confirm the information above is accurate and I accept the Terms of Service and Privacy Policy.
HAEMA
Every drop tells a story. We connect donors, patients, and hospitals so no one waits for the blood that could save them.
Quick Links
Donate Blood
Request Blood
Find a Camp
Schedule Checkup
Blood News
Support
Contact Us
About Us
FAQs
Privacy Policy
Terms & Conditions
© 2026 HAEMA. All rights reserved.
Donate Blood / Save Lives

Fig.11

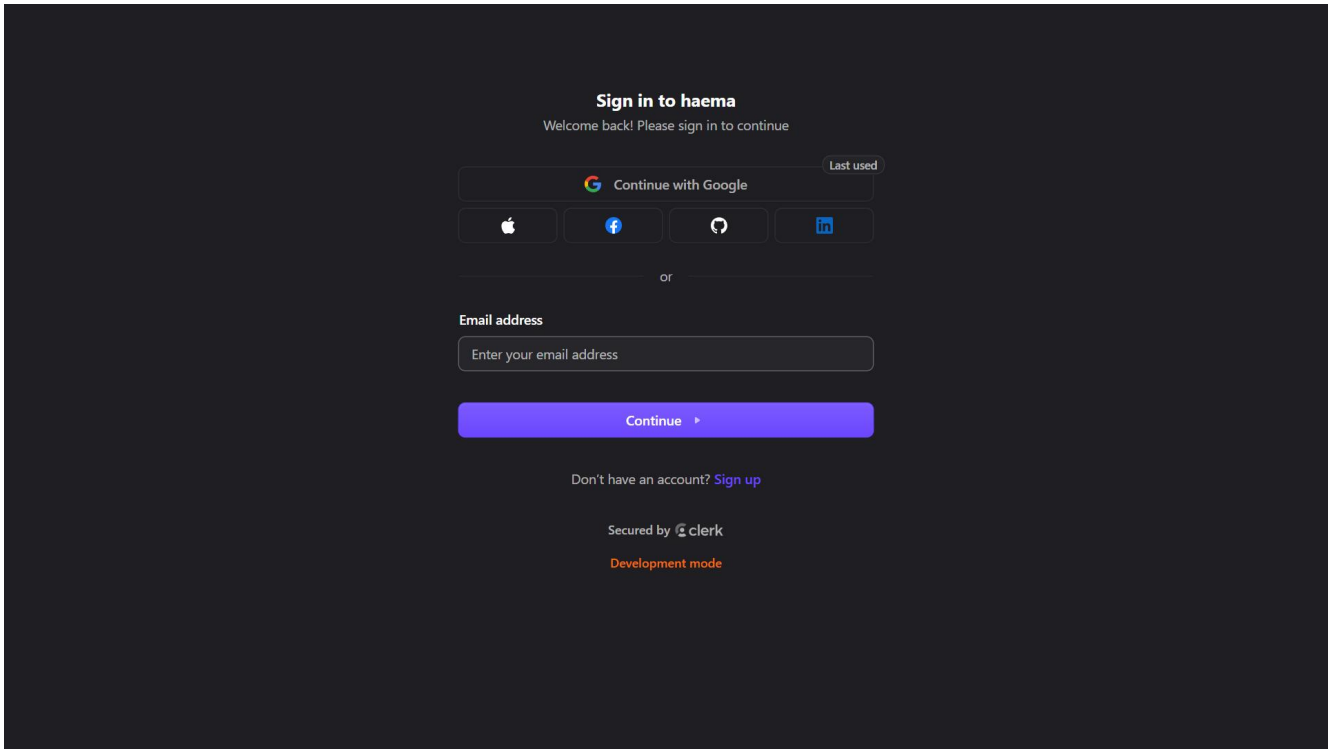
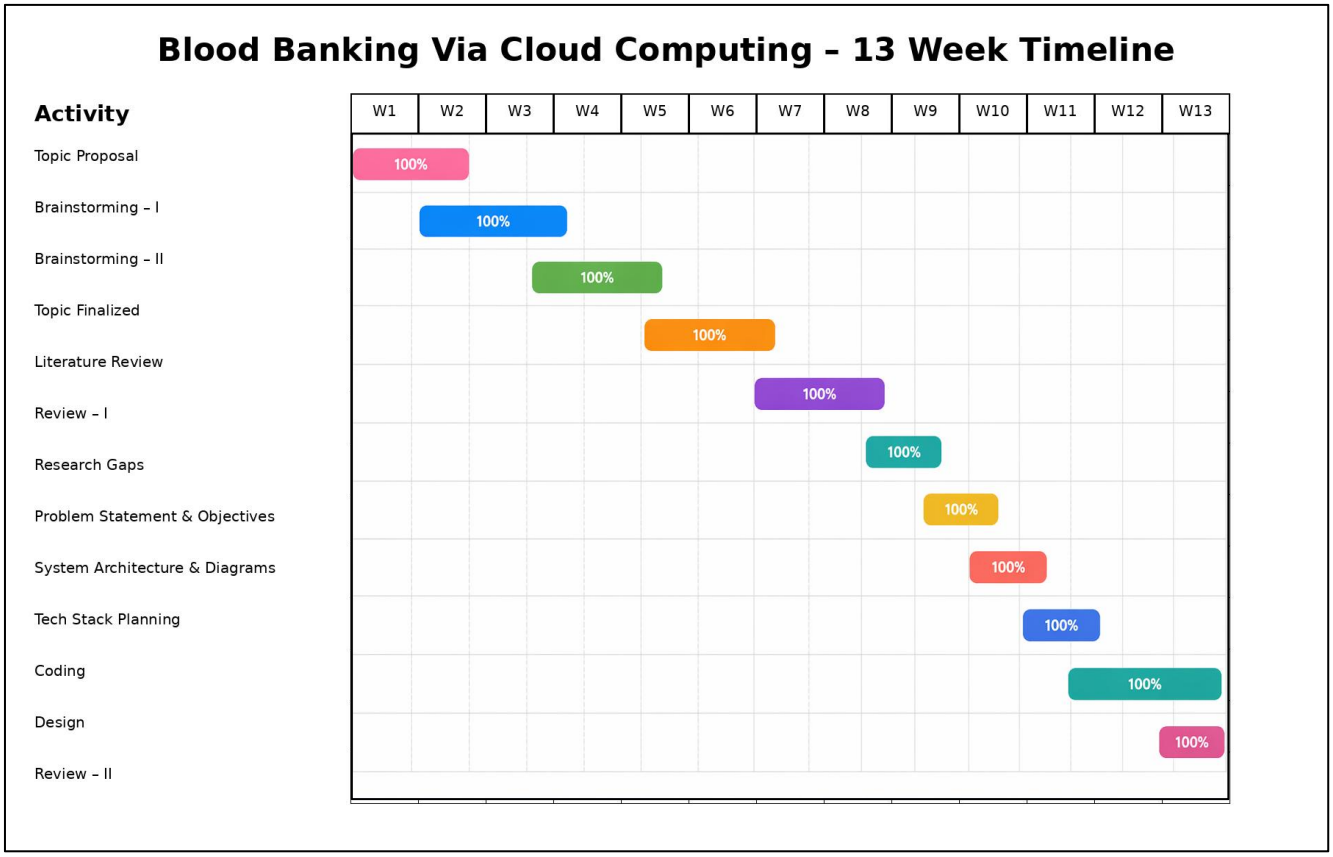

Sign in to haema
Welcome back! Please sign in to continue
Last used
Continue with Google
or
Email address
Enter your email address
Continue
Don't have an account? Sign up
Secured by clerk
Development mode

Fig.12

TIMELINE CHART



CONCLUSION

The **Blood Banking Via Cloud Computing** project proposes a centralized digital solution for managing blood-bank operations. By utilizing cloud computing, the system can provide improved accessibility, centralized data management, easier record maintenance, and better scalability compared with conventional approaches.

The proposed system covers important functions such as **donor management, blood inventory management, blood searching, blood requests, authentication, and administration**. The planned architecture separates the presentation, application, and database layers, making the system easier to maintain and extend.

In the future, the system can be enhanced with **real-time notifications, emergency blood requests, location-based blood-bank searching, SMS/email alerts, analytics, mobile applications, and intelligent blood-demand prediction**.

FUTURE SCOPE

The **Blood Banking Via Cloud Computing** system can be further enhanced with advanced technologies and additional features to make blood management faster, more reliable, and accessible.

1. Mobile Application

A dedicated Android/iOS application can be developed so that donors, hospitals, and administrators can access the system through smartphones.

2. Real-Time Blood Availability

The system can provide **real-time blood-stock information** by automatically updating inventory whenever blood is collected, issued, or transferred.

3. Emergency Blood Request

An emergency-request feature can allow hospitals or patients to mark a request as urgent and notify nearby eligible donors or blood banks.

4. Location-Based Blood Bank Search

GPS and location services can be integrated to help users find the **nearest blood bank or available donor** based on their location.

5. SMS and Email Notifications

The system can send automatic notifications for:

- Blood requests
- Donation reminders
- Request approval/rejection
- Low blood-stock alerts
- Emergency requirements

6. AI-Based Blood Demand Prediction

Machine Learning can be incorporated to analyze historical blood usage and predict future demand for different blood groups. This can help blood banks maintain adequate inventory.

7. Cloud-Based Analytics Dashboard

An analytics dashboard can display:

- Total donors
- Blood collected
- Blood issued
- Current inventory
- Most requested blood groups
- Monthly donation trends

8. Improved Security

Advanced security mechanisms such as **multi-factor authentication, encryption, role-based access control, and audit logs** can be implemented to protect sensitive healthcare information.

9. Integration with Hospitals

The platform can be integrated with hospitals and healthcare organizations so that authorized hospitals can directly submit and track blood requests.

10. IoT-Based Blood Storage Monitoring

IoT sensors can be used to monitor storage conditions such as **temperature and humidity** and generate alerts if conditions become unsafe.

11. Digital Donor History

A complete digital donation history can be maintained for each donor, helping authorized personnel track previous donations and donor eligibility.

12. Scalable Multi-Blood-Bank Network

In the future, multiple blood banks from different locations can be connected to the same cloud platform, creating a **centralized blood availability network**.

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